TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Determination of the Coefficient of Viscosity of a Liquid by Its Flow through a Capillary Tube

Poiseuille's law at the bench — a constant head drives the liquid through a fine bore, and the timed flow rate yields η, with its unforgiving fourth power of the radius.

PHY-1104 · PHYSICS-I SESSIONAL · EXPERIMENT 8 · LEVEL 1 TERM 1

1. Aim and theory

Aim: to find the coefficient of viscosity η of a liquid (water in the model run) from its rate of flow through a capillary tube. For streamline (laminar) flow through a tube of radius r and length L under a pressure difference P, Poiseuille's law gives the volume flow rate

$$Q = V/t = \pi P r^4 / (8 \eta L) \;\Rightarrow\; \eta = \pi P r^4 t / (8 L V)$$

The pressure is supplied by a constant liquid head h: P = ρgh, maintained by a Mariotte bottle or an overflow arrangement so that h — and therefore Q — stays steady during a run. The law's conditions are the experiment's conditions: the flow must be **laminar** (slow, in a fine tube — Physics-I Topic 5), the tube's bore uniform, and end effects negligible (L ≫ r).

2. Apparatus and procedure

- **Apparatus:** capillary tube mounted horizontally, constant-head reservoir, measuring cylinder and stopwatch, travelling microscope or mercury-thread measurement for the bore, thermometer, metre rule.
- Measure the tube's length L and bore radius r (mercury thread method, as in Experiment 6).
- Set a head h; open the flow and let it run until the rate is steady. Collect the outflow in the measuring cylinder for a timed interval; record V and t; compute Q = V/t.
- Repeat for several heads h; Q should be proportional to h (the laminar check). Compute η from each run and average — or take η from the slope of the Q–h graph.
- Record the liquid's temperature: η of water changes by about 2% per °C, so a temperature is part of the result.

3. Worked calculation (model data)

RESULT

Tube: L = 0.200 m; r = 0.50 mm = 5.0×10^{-4} m → $r^4 = 6.25 \times 10^{-14}$ m⁴. Head h = 0.300 m of water → $P = \rho gh = 1000 \times 9.8 \times 0.300 = 2940$ Pa.

Run: V = 50.0 mL = 5.00×10^{-5} m³ collected in t = 140 s.

$\eta = \pi P r^4 t/(8LV) = (\pi \times 2940 \times 6.25 \times 10^{-14} \times 140)/(8 \times 0.200 \times 5.00 \times 10^{-5}) = (8.08 \times 10^{-8})/(8.00 \times 10^{-5})$ = **1.01×10^{-3} Pa·s**.

Water's accepted η at 20 °C is 1.002×10^{-3} Pa·s — the model run (at 20 °C) agrees. Laminar check: mean velocity in the tube = $Q/(\pi r^2) = (5.00 \times 10^{-5}/140)/(\pi \times 2.5 \times 10^{-7}) = 0.455$ m s⁻¹; Reynolds number = $\rho v(2r)/\eta = 1000 \times 0.455 \times 0.0010/0.00101 \approx 450$ — well below the ~2000 threshold, so Poiseuille's law was entitled to apply.

4. Precautions

- The head must be genuinely constant during a timed run — use the overflow/Mariotte arrangement and read h at the tube's axis level.
- Measure r with the same care as in the rigidity experiment: $\eta \propto r^4$, so 1% in r is 4% in η.
- Let the flow become steady before starting the clock; discard the first runnings.
- Keep the flow slow enough to stay laminar; if Q stops being proportional to h, turbulence has begun and those runs are invalid.
- Note the temperature and compare only with the accepted η at that temperature.
- The tube must be horizontal and clean; a tilted tube adds or removes hydrostatic head along its length.

5. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. State Poiseuille's law and the two flow conditions it requires.
2. For the model apparatus, how long would 50.0 mL take at a head of 0.200 m instead of 0.300 m?
3. Why is a constant-head device used instead of simply letting a tank drain?
4. A second tube has twice the radius, same length. At the same head, how does Q change?
5. Why does the Reynolds-number check belong in the write-up?
6. The measured η comes out 5% high on every run. Name the measurement most likely responsible, and the check you would run.

Answer key

1. $Q = \pi P r^4/(8\eta L)$: the volume per second through a capillary under pressure difference P. It requires laminar (streamline) flow and a long, uniform tube ($L \gg r$) so that entrance effects are negligible and the velocity profile is fully developed.
2. Q is proportional to P, hence to h: at 2/3 of the head, Q falls to 2/3, so the same volume takes 3/2 as long: $140 \times 1.5 = 210$ s.
3. Because a draining tank's head — and thus P and Q — falls continuously during collection, mixing many rates into one timing. The constant-head device holds h fixed, so the measured Q is a single, definite rate belonging to a single P.
4. $Q \propto r^4$: doubling r multiplies Q by 16. (The same fourth power that makes bore measurement the experiment's critical step.)
5. Because Poiseuille's law is a laminar-flow result. Computing Re for the run proves the regime the law assumes actually held ($Re \approx 450 \ll 2000$ in the model); without the check, a turbulent run would silently return a meaningless "η".
6. The bore radius r — $\eta \propto r^4$ means a mere 1.2% underestimate of r inflates η by 5%. Check: re-measure the bore by the mercury-thread method at several positions along the tube (a tapered bore is the usual culprit) and use the properly averaged r^4.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib